Issue	**01**
Factory ID	**25401**
Date of Issue	**03-Jul-2025**

Velocity Textiles

F-235, S.I.T.E, Near Indus University, (Site Campus) 75700, Karachi
Karachi
Sindh Pakistan

Latitude: [24.905947], longitude: [66.988821]
Date of inspection: 29-Jan-2025

Structural Safety Inspection Report

Observations & Actions



Authors	**Muneeba Ashraf**
Reviewed by	**Tauqir Ahmad**
Approved by	**Paul Rigby**

Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 29-01-2025 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

Contents

1. Executive Summary 4
2. Building Extents 6
   2.1 Structural System 15
3. Documentation 21
4. Structural Safety Measures 22
5. Cracking 24
   5.1 Column 24
6. Other Damage/Defects 25
   6.1 Spalling concrete 25
   6.2 Spalling concrete 27
   6.3 Column impact damage 29
   6.4 Health and Safety 30
7. Structural Documentation 31
   7.1 Documentation unavailable 31
   7.2 Non-engineered structure 32
   7.3 Lack of safe load plans 34
8. Assessment Limitations and Assumptions 35

1. Executive Summary

On 29-Jan-2025, Dr. Aslam Faqeer Mohammad and Muneeba Ashraf on behalf of the Pakistan Accord carried out an initial structural inspection for Velocity Textiles, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representatives Mr. Rizwan Butt (Admin Manager), Mr. Syed Mudassir (Compliance officer), Mr. Hamza (Compliance officer), Mr. Uzair (Electrical engineer) and other representatives from the factory were also present. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were also present.
The factory operates one general shift. Approximately 520 workers are employed in the factory, including 80-90 female workers.
The factory mainly comprises of 02 buildings and utilities, tenanted by Velocity Textiles. During the inspection, the area highlighted in Figure 1.1 was inspected, with the site boundary indicated in red. One building and a scrap area were not inspected, as they are shared with other tenants. While the area not inspected during the initial inspection will be covered in the follow-up inspection.

- **Building 1,** a five-storey (Ground+Mezzanine+3) is a combination of reinforced concrete moment resisting frame (RC MRF) and composite slab system with masonry infill on periphery, roof is vertically extended with non-engineered steel shed, is used for offices and washing (ground floor), stitching (Mezzanine floor), stitching, checking and packing (1st and 2nd floor) and storage of goods, prayer area (3rd floor) and solar panels (roof). The building was constructed in 2000.
- **Building 2,** a two-storey (Ground+1) is a reinforced concrete moment resisting frame (RC MRF) with masonry infill and roof is of steel portal frame, is used for cutting, storage, warehouse and chemical store (ground), stitching (1st floor) and solar panels (roof). The year of construction is 2000.
- **Utilities,** ground only four different structures which include boiler shed, generator rooms, LT rooms and reinforced concrete ETP structure and a compressor is installed on the roof. The year of construction is unknown.

We were presented with approved site layout plan of the factory and stability certificate of Building 1 and 2 and other documents listed in table 2.
A summary of the main structural concerns identified during the inspection includes (non-exhaustive):

- Cracking of columns at 2nd floor was observed in building 1.
- A damaged column at ground floor was observed in building 2.
- Spalling of slab concrete at 2nd floor and projection at all levels in building 1 and at ground floor in building 2.
- Presence of non-engineered steel shed
- Lack of documentation and safe load plans.

Two Health and Safety issues related to the lack of edge protection at the compressor area and using asbestos in prayer and childcare areas. Edge protection should be addressed immediately.
We see no reason to suspend operations in the facility due to these concerns (subject to the required actions noted in this report being addressed

within agreed timeframes).
Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

Table 1 - Building Summary

Building	Use	No. of stories	Year of construction	Structural system	Roof structural system	Any Extension/ modifications	Engineer access issues
Building 1	Offices and washing (ground floor) Stitching (Mezzanine floor) Stitching, checking and packing (1st and 2nd floor) Storage of goods, prayer area (3rd floor) Solar panels (roof).	Ground+Mez+3 (five-story)	2000	Reinforced Concrete Moment resisting frame 2nd floor and composite slab system from ground to 1st floor.	Partially covered with non-engineered steel shed	Non-engineered steel shed	No

Building 2	Cutting, storage, warehouse and chemical store (ground) Stitching (1st floor) Solar panels (roof).	Ground+1 (two-story)	2000	Reinforced Concrete Moment resisting frame	Steel portal frame	No	No
Utilities	Boiler shed, generator rooms, LT rooms	Ground only	Unknown	Reinforced Concrete Moment resisting frame	-	No	No



Figure 1.1. Site plan (Satellite view) - Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected.



Figure 1.2. Site Layout of velocity textiles.



Figure 1.3. Building 1, Front elevation



Figure 1.4. Building 1, side elevation



Figure 1.5. Building 1 and utilities, side elevation



Figure 1.6. Building 2, Side elevation



Figure 1.7. Building 1, Storage of goods at roof



Figure 1.8. Building 1, Checking at 2nd floor



Figure 1.9. Building 1, stitching at 2nd floor



Figure 1.10. Building 1, Typical Stitching activity at 1st floor



Figure 1.11. Building 1, Typical stitching activity at Mezzanine floor



Figure 1.12. Building 1, Washing activity at ground floor



Figure 1.13. Building 2, Solar panels on roof



Figure 1.14. Building 2, Typical Stitching activity at 1st floor



Figure 1.15. Building 2, Sample stitching at Ground floor



Figure 1.16. Building 2, Fabric storage at Ground floor



Figure 1.17. Building 2 extended shed, storage at Ground floor



Figure 1.18. Utilities, wood boiler



Figure 1.19. Utilities, LT room



Figure 1.20. Utilities, Generator room



Figure 1.21. Utilities, ETP

2.1 Structural System

Building	Building 1
Stability system	A=non-engineered steel shed B=non-engineered steel shed
Column size (m)	-
Grid (m)	-
Primary beam size (m)	-
Secondary beam size (m)	-
Slab depth (m)	-
Finishes depth (m)	-
Ceiling heights (m)	A=3.75 B=4.15

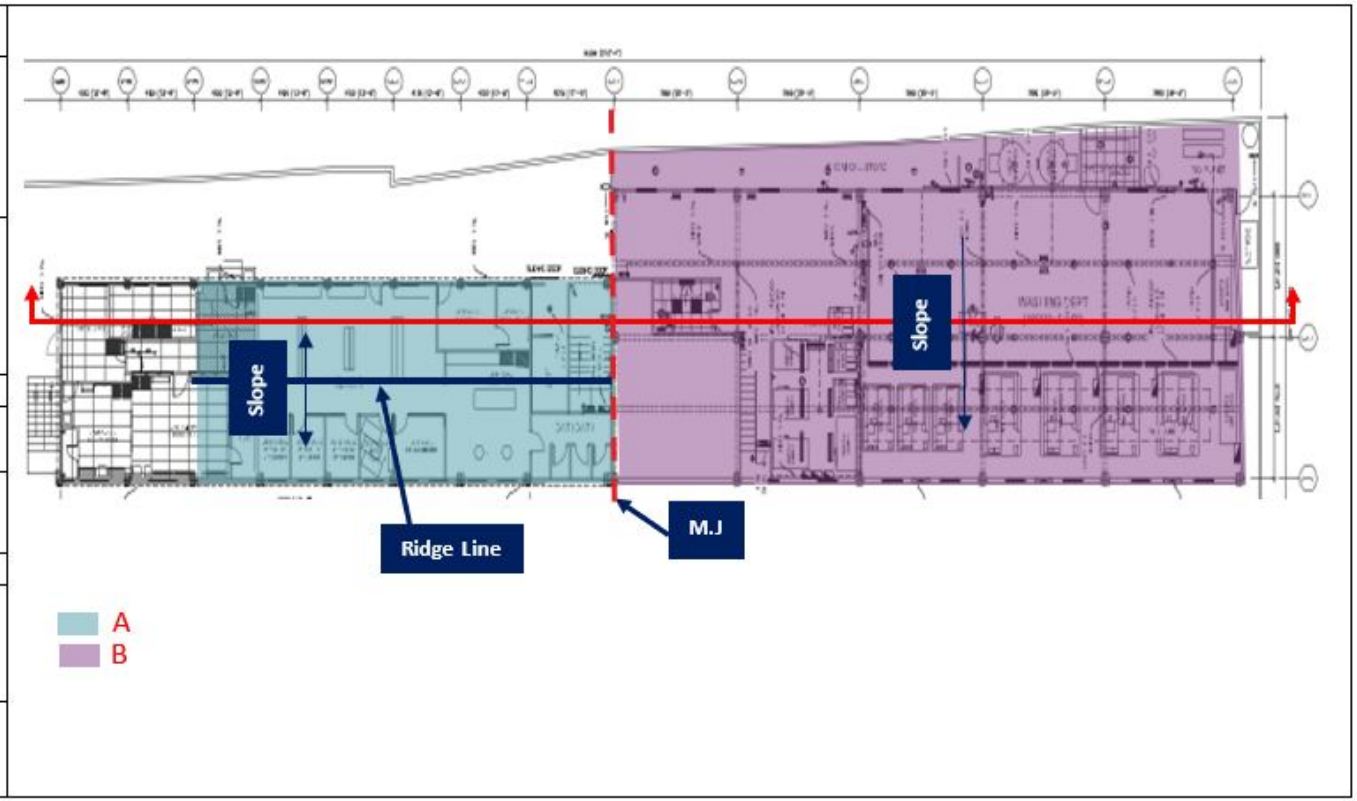


Figure 1.22. Roof framing plan of Building 1

Building	Building 1
Stability system	A=RC MRF & Composite slab B= RC MRF
Column size (m)	0.48x0.48 (A&B)
Grid (m)	3.81x7.6 (A) 5.4x7.16 (B)
Primary beam size (m)	0.3x0.61 (A) 0.3x0.6 (B)
Secondary beam size (m)	0.3x0.61 (A) 0.3x0.6 (B)
Slab depth (m)	A=0.125 Typ. B=0.2 (2nd floor) B=0.125 (1st floor)
Finishes depth (m)	0.075
Ceiling heights (m)	A=3.64 (Ground floor) A=2.56 (Mezzanine floor) A=3.64 (1st floor) A= 3.4 (2nd floor) B=4.7 (Ground floor) B=4.2 (1st floor) B=3.96 (2nd floor)

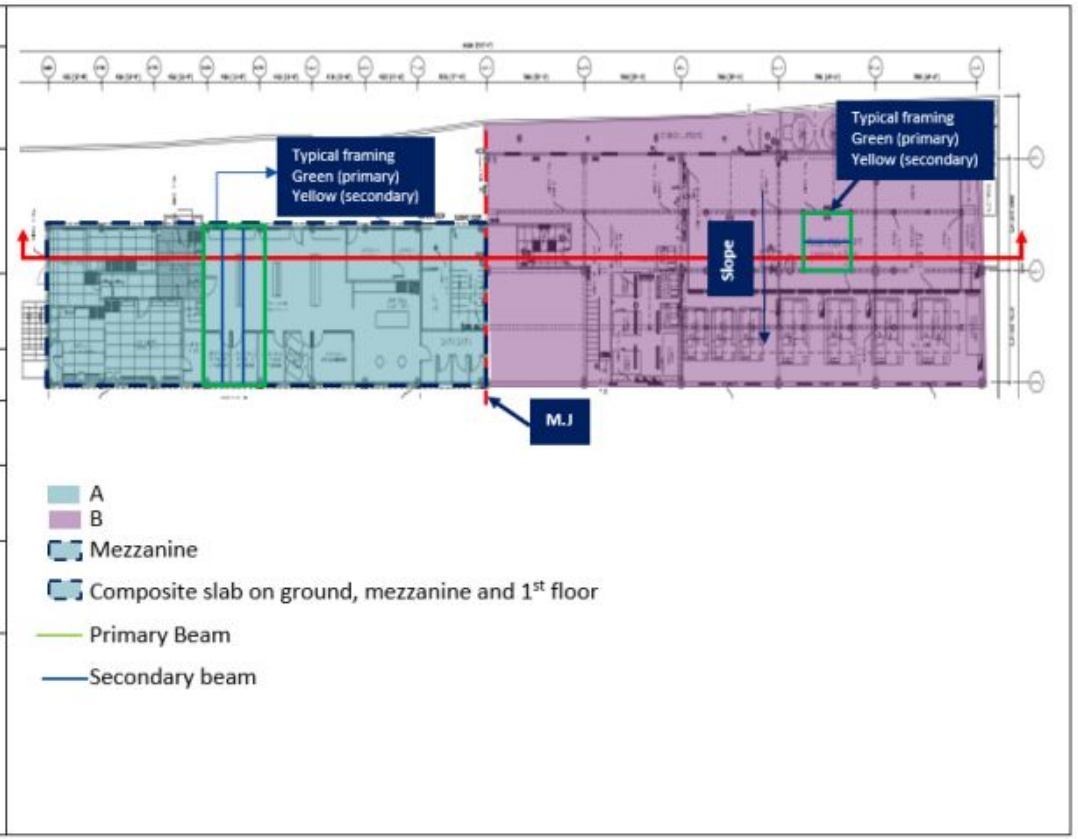
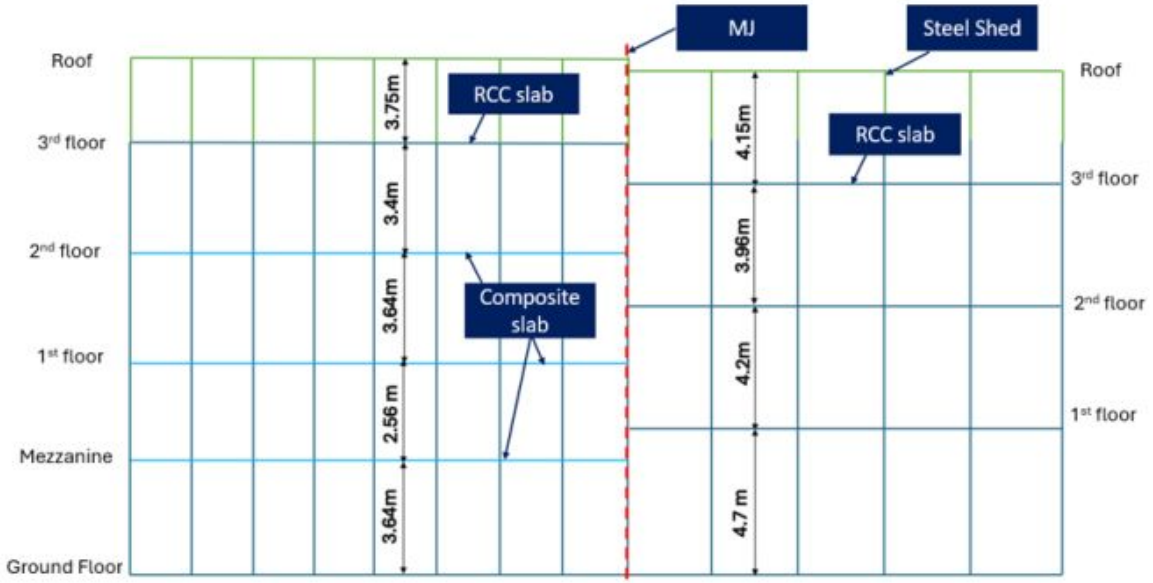


Figure 1.23. Typical framing plan of Building 1



Structural System Notes

Column size is the structural dimension and excludes render thickness

Grid measurements are from the centre of column to centre of column

Beam sizes do not include slab depth

Figure 1.24. Typical section of Building 1

Building	Building 2
Stability system	Steel portal frame
Column size (m)	-
Grid (m)	7.0x7.0
Primary beam size (m)	-
Secondary beam size (m)	-
Slab depth (m)	-
Finishes depth (m)	-
Ceiling heights (m)	5.0

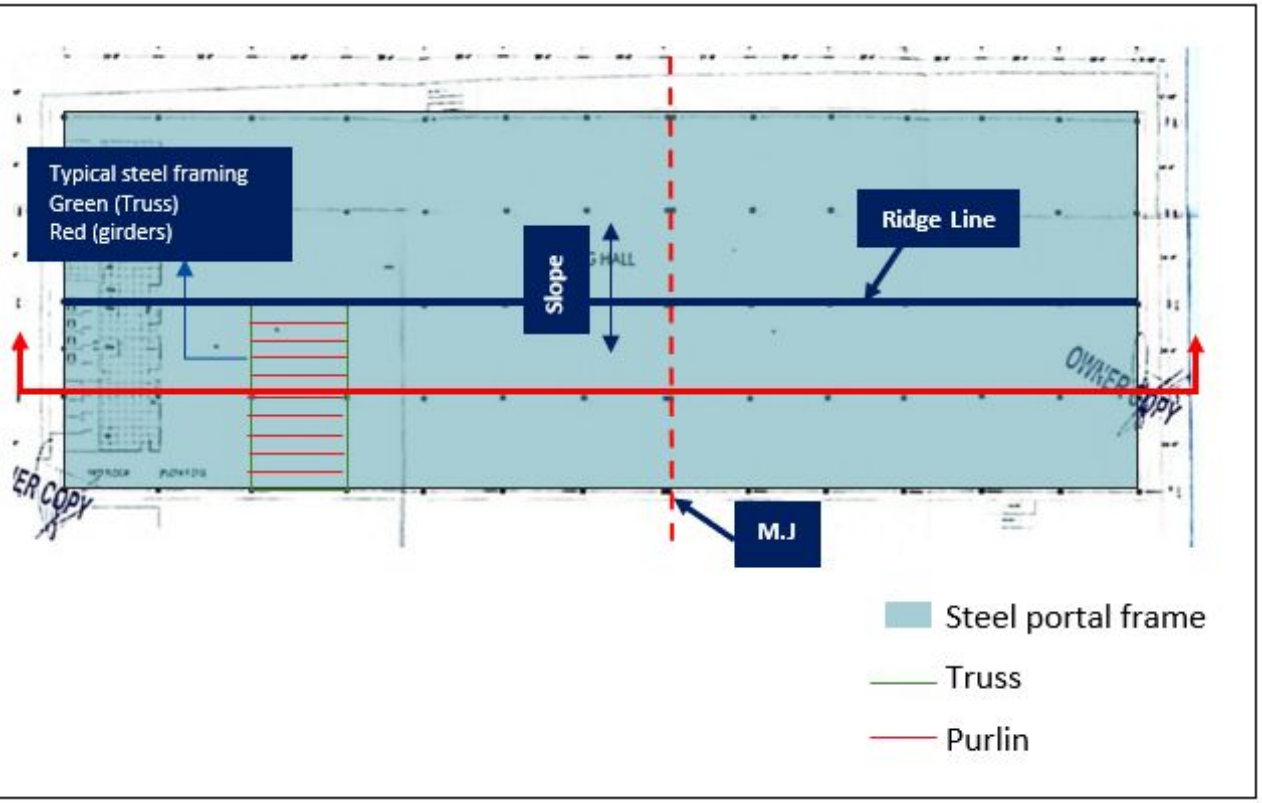


Figure 1.25. Roof framing plan of Building 2

Building	Building 2
Stability system	RC moment resisting frame
Column size (m)	0.4x0.4
Grid (m)	7.0x7.0
Primary beam size (m)	0.4x0.76
Secondary beam size (m)	0.2x0.46 0.4x0.76
Slab depth (m)	0.15
Finishes depth (m)	0.1
Ceiling heights (m)	5.4

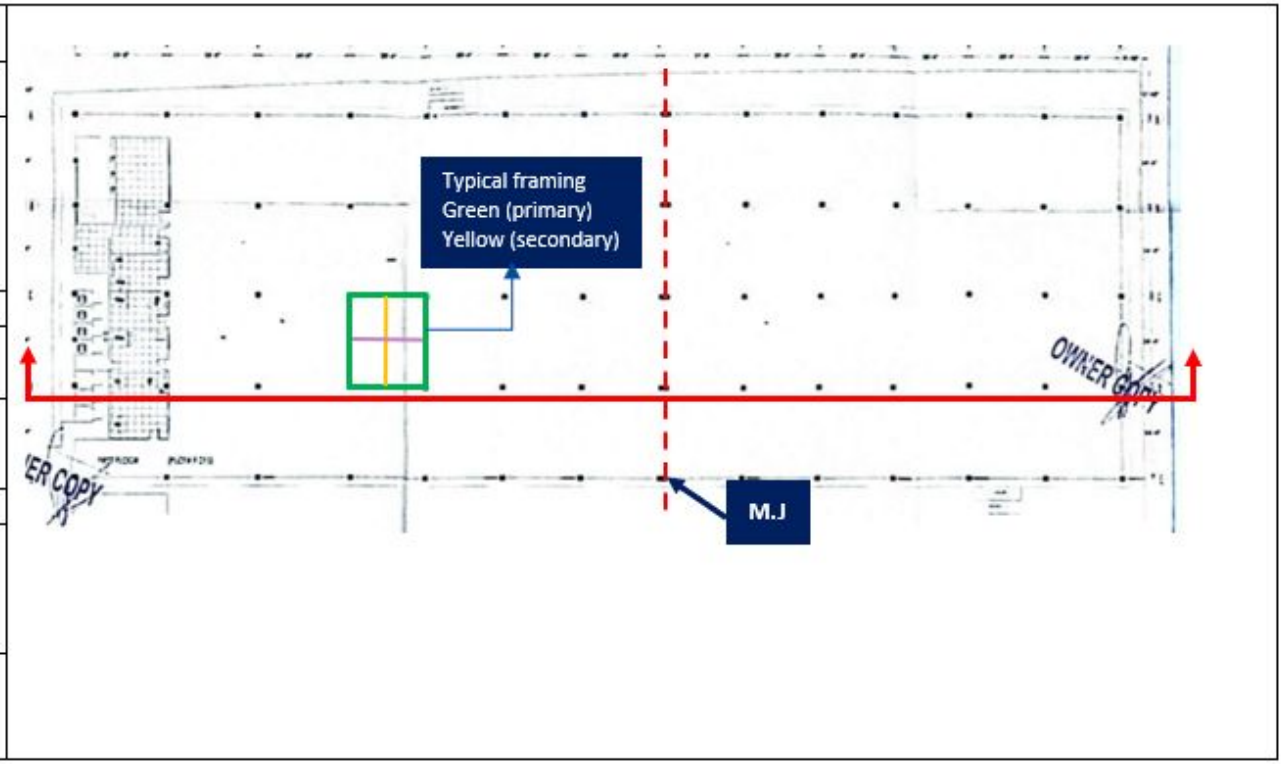
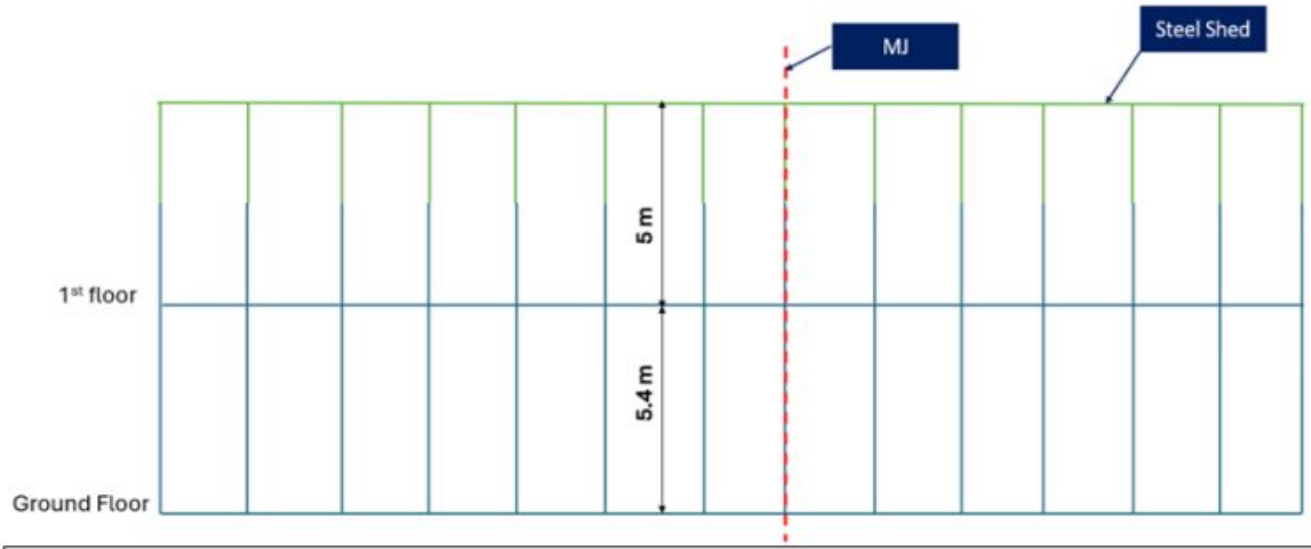


Figure 1.26. 1st floor framing plan of Building 2



Structural System Notes

Column size is the structural dimension and excludes render thickness

Grid measurements are from the centre of column to centre of column

Beam sizes do not include slab depth

Figure 1.27. Typical section of Building 2

3. Documentation

We were presented with the following design documentation. The site layout and approved plans of the buildings were available.

Table 2 Documentation

Building	Site layout	Approved layout/permit	Geotech report	Design drawings	As-built drawings	Loading plan	Stability certificate	Inspection reports	Materials testing
Building 1	Yes	Yes	No	No	No	No	Yes	No	No
Building 2	Yes	Yes		No	No		Yes	No	No
Utilities	Yes	Yes		No	No		No	No	No

4. Structural Safety Measures

During our structural site inspection of Velocity Textiles, a number of structural issues were identified.

It is essential to address these structural safety issues promptly to enhance safety and compliance within Velocity Textiles. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

Table 3 Summary of issues by building:

Building	Potential Overstress	Structural system	Cracking	Loading	Other damage and defects	Structural documentation
Building 1	-	-	Column	-	Spalling of Concrete Non-engineered Structure	Documentation Unavailable Lack of Safe Load Plans
Building 2	-	-			Spalling of Concrete Column impact damage	Documentation Unavailable Lack of Safe Load Plans

Utilities	-	-	-	-	Lack of Edge protection	Documentation unavailable

5. Cracking

5.1 Column

This issue refers to the 2nd floor of building 1, where a single column cracking was observed. The potential cause of cracking might be corrosion of reinforcement or water ingress. However, the column is cleared in stress check analysis. F-202503-014841

Timeline	Remediation Plan
Medium Term FIN-0017423	Structural engineer to carry out a detailed assessment of the column to identify the cause of the cracking. Propose measures to prevent future damage and to repair the existing damage and submit to Pakistan Accord for approval.
Longer Term FIN-0017424	Carry out the remedial works.



Figure 2.1 - Cracking of column at 2nd floor of Building 1

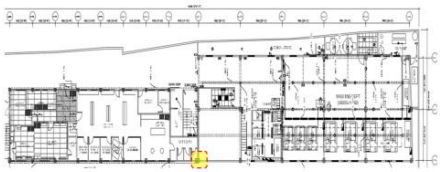


Figure 2.2 - Location for cracking of column in Building 1

6. Other Damage/Defects

6.1 Spalling concrete

This issue refers to building 2. Spalling of concrete at the ground floor and projected slab was observed. The potential cause of spalling might be due to lack of waterproofing. F-202503-014845

Timeline	Remediation Plan
Medium Term FIN-0017521	Identify the cause of the damage and propose measures to prevent further water damage Inspect damage to the slab and propose repairs to structure. Submit report capturing above to Pakistan Accord for approval.
Longer Term FIN-0017522	Implement works Continue to monitor for future signs of water ingress/damage



Figure 2.3 - Signs of concrete spalling at ground floor



Figure 2.4 - Signs of concrete spalling of projection



Figure 2.5 - Location of spalling on the plan

6.2 Spalling concrete

This issue refers to building 1. Spalling of concrete at 2nd floor and projection slab was observed. Some areas were repaired earlier, and few small areas were about to spall at 2nd floor. F-202503-014846

Timeline	Remediation Plan
Medium Term FIN-0017526	Identify the cause of the damage and propose measures to prevent further water damage Inspect damage to the slab and propose repairs to the structure. Submit report capturing above to Pakistan Accord for approval.
Longer Term FIN-0017527	Implement works Continue to monitor for future signs of water ingress/damage



Figure 2.6 - Signs of concrete spalling at 2nd floor

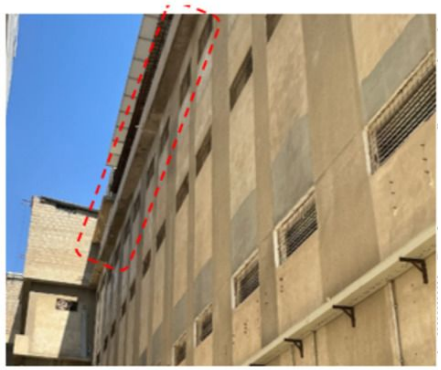


Figure 2.7 - Signs of repaired spalling



Figure 2.8 - Location of spalling at 2nd floor on the plan

6.3 Column impact damage

This issue refers to the ground floor of building 2, where a single column was found to be damaged with exposed reinforcement. The potential cause of the damage appears to be impact from trolleys. F-202503-014847

Timeline	Remediation Plan
Medium Term FIN-0017528	Repair damage to column
Longer Term FIN-0017529	Install protective measures to prevent damage recurring Continue to monitor the condition of columns in highly trafficked area.



Figure 2.9 - Damaged column at ground floor of Building 2

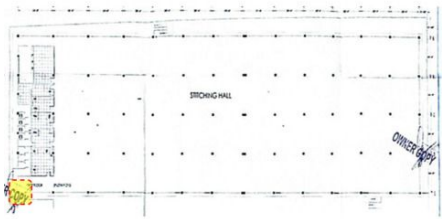


Figure 2.10 - Location of damaged column at ground floor of Building 2

6.4 Health and Safety

This concerns inadequate edge protection in utilities on roof where compressor has been placed but inadequate edge protection. This creates a risk of injury resulting from trip or fall – Health & Safety concern. F-202504-015592

Timeline	Remediation Plan
Immediate FIN-0019087	Provide adequate edge protection on roof. Ensure similar measures are implemented in any other factory areas where these safety concerns exist.



Figure 2.11 - Inadequate edge protection at roof of the utility building

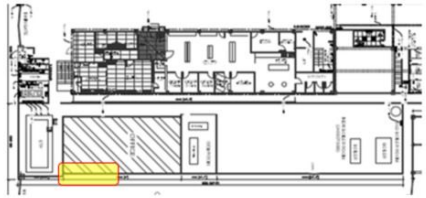


Figure 2.12 - Indicates inadequate edge protection at roof of utility building

7. Structural Documentation

7.1 Documentation unavailable

This issue refers to all buildings.
The architectural and structural drawings are unavailable and need to be prepared. F-202503-014843

Timeline	Remediation Plan
Longer Term FIN-0017518	Building Structural Engineer to carry out on-site survey, collect information and produce accurate and complete as-built structural drawings.
Longer Term FIN-0017520	Building Structural Engineer to carry out on-site survey, collect information and produce accurate and complete as-built structural drawings.

7.2 Non-engineered structure

Non-engineered steel shed structures observed in Building-1. These structures might be unstable against gravity and wind loads. F-202504-015591

Timeline	Remediation Plan
Medium Term FIN-0019085	Building structural engineer to review the as-built design for code compliance (vertical and lateral loads) and confirm if it is adequate. Propose strengthening works as required and submit to Pakistan Accord for approval.
Longer Term FIN-0019086	Implement strengthening works as required



Figure 2.13 - Non-engineered shed at roof of building 1



Figure 2.14 - Non-engineered shed at roof of building 1



Figure 2.15 - Location of non-engineered shed at roof of building 1

7.3 Lack of safe load plans

No safe load plans were posted all over the factory. F-202503-014844

Timeline	Remediation Plan
Longer Term FIN-0017519	Building structural engineer to review design documentation to confirm load capacity of the structure. Develop safe load plans and submit to Pakistan Accord for review. Factory to post safe load plans on each floor.

8. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.